The permutation p can be represented as a collection of **disjoint cycles**. Therefore, we can solve each cycle independently.

Consider a cycle of length k:

$$C = \{v_1, v_2, \ldots, v_k\}$$

If we make the **entire cycle happy**, exactly $k-1$ swaps are required.
Hence, the profit is:

$$P_{\text{all}} = \sum_{i \in C} a_i - (k-1)c$$

Now suppose we choose an index j and make person j happy individually.
The cost of the swap is c, while the reward is a_j. Therefore, its net profit is:

$$a_j - c$$

After fixing j, this element becomes disconnected from the remaining cycle, so we can repeat the same process on the remaining elements.

Therefore, if we independently choose some elements to fix, the contribution of an element i is:

$$\max(0, a_i - c)$$

Thus, the profit when we **do not necessarily fix the whole cycle** is:

$$P_{\text{individual}} = \sum_{i \in C} \max(0, a_i - c)$$

However, there is an important advantage when we fix the **complete cycle**: k people can be made happy using only $k-1$ swaps.
So we calculate both possibilities:

$$P_{\text{individual}} = \sum_{i \in C} \max(0, a_i - c)$$

$$P_{\text{all}} = \sum_{i \in C} a_i - (k-1)c$$

The best profit for this cycle is:

$$\boxed{P_C = \max(P_{\text{individual}}, P_{\text{all}})}$$

For example, suppose:

$$C = \{j_1, j_2, \ldots, j_k\}$$

If $a_{j_1} < c$, fixing j_1 individually gives negative profit:

$$a_{j_1} - c < 0$$

So we normally avoid fixing it individually.

Similarly, for every j such that $a_j < c$, its individual contribution is 0:

$$\max(0, a_j - c) = 0$$

But sometimes several such elements together can make the **complete-cycle option** better because completing the cycle costs only $k-1$ swaps instead of k.

For example, if two elements satisfy:

$$a_{j_1} < c, \qquad a_{j_2} < c$$

We should compare:

$$a_{j_1} + a_{j_2}$$

with the additional swap cost saved by completing the cycle.

This is exactly why we compare $P_{\text{individual}}$ with P_{all}, rather than handling every subset separately.

For a cycle of length 1, the person already has their own token, so:

$$P_C = a_i$$

Finally, if the permutation contains cycles:

$$C_1, C_2, \ldots, C_m$$

The total answer is:

$$\boxed{\text{Answer} = \sum_{r=1}^{m} P_{C_r}}$$

We find all cycles using DFS and a `visited` array.

Every vertex belongs to exactly one cycle, so every vertex is processed only once.

Therefore:

$$\boxed{\text{Time Complexity} = O(n)}$$

$$\boxed{\text{Space Complexity} = O(n)}$$